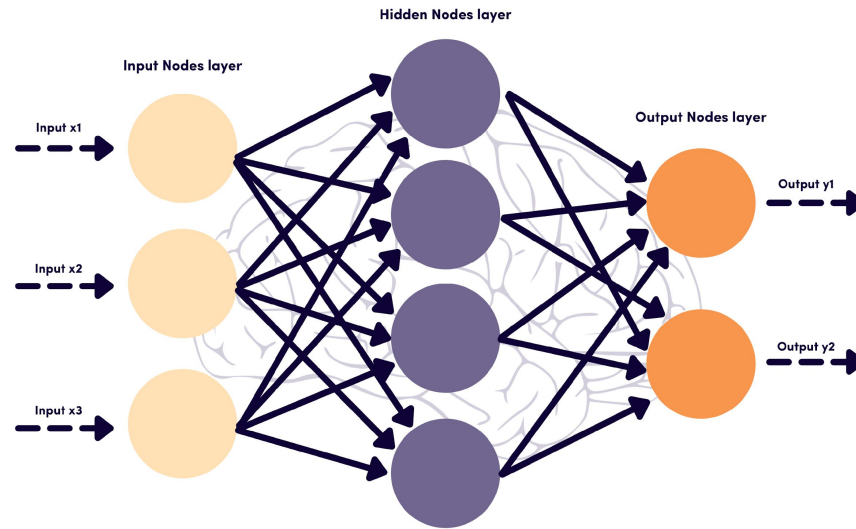


Exercise session on ANN

Is it Daisy?

ANN

ANNs are made up of layers of neurons, where each neuron in a layer is connected to every neuron in the previous and next layers. The data flows through these layers sequentially, with each layer transforming the input data using a weighted sum and an activation function.



ANNs are more general-purpose and can be used for various tasks, such as:

Tabular Data Classification/Regression: Predicting outcomes based on structured data (e.g., predicting house prices based on features).

Time-Series Forecasting: Predicting future data points in a time series.

Basic Image and Text Processing: They can be used for simple image and text classification tasks, but they are not optimized for these types of data.

Input Data:

Typically expects input data in a 1D vector format. For images, this means flattening the 2D image into a 1D vector.

Advantages:

Simple and easy to implement for a wide range of tasks. Less computationally intensive compared to CNNs.

Limitations:

Not well-suited for image data, as they do not naturally capture spatial hierarchies in images. Prone to overfitting, especially on high-dimensional data like images.